

Dr. Ellen Leitinger

Research fellow in Astrophysics

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Research positions and education

DIFA Unibo & INAF-OAA, Italy

Research Fellowship — Since 2024

Member of the ERC funded **'Asterochronometry'** and **'StarDance'** projects

Galactic archaeology | Asteroseismology | Binary stars

UQ, Australia & ESO, Germany

Ph.D. in Astrophysics — Graduated 2025

Title: 'A wide-field view of multiple stellar populations in 30 Galactic globular clusters'
Supervisors: A/Prof. Holger Baumgardt, Dr. Michael Hilker and Dr. Ivan Cabrera-Ziri

Globular clusters | Multiple stellar populations | Photometry | Stellar kinematics and dynamics

UQ, Australia

Masters Degree in Astrophysics

Title: 'Contributions of the MgII emission line in reverberation mapping of active galactic nuclei'
Supervisors: Dr. Janie Hoorman and Prof. Tamara Davis
Thesis Grade: High Distinction [7/7]

AGN | Reverberation mapping | MCMC methods | Spectroscopy | Time series analysis

Research projects

Fellowship

Nitrogen-rich field stars through the lens of asteroseismology

I identified a sample of metal-poor Galactic field stars with chemical abundance patterns consistent with formation within globular clusters, using APOGEE abundances and *Gaia* astrometry. Following up with asteroseismology from the *Kepler* and K2 missions, I discovered that most of the enriched stars are too young to have originated in globular clusters, challenging previous assumptions of their origins.

PhD

A wide-field view on multiple stellar populations in 30 Milky Way globular clusters

Through the homogeneous analysis of *HST* and Johnsons-Cousins photometry with a large sample of 30 Galactic globular clusters, I photometrically classified the multiple stellar populations within each cluster and determined their spatial distributions. I discovered that dynamically young GCs display a wide range of initial conditions, which challenged previous formation theories. I then combined proper motions and line-of-sight velocities to investigate the 3D kinematics. I determined the rotation, velocity dispersions and anisotropy of all stars in each cluster, as well as the individual populations.

Masters

The contribution of the MgII emission line in reverberation mapping of AGN

Reverberation mapping is a common method for determining the masses of AGN using multi-epoch data, most commonly with the use of the Hβ emission line. I determined that the same analysis could be completed for higher redshift AGN $1 < z < 3$ using the MgII line, despite contamination from the surrounding Fe emission lines.

First author publications

- 'A Wide-Field View on Multiple Stellar Populations in 28 Milky Way Globular Clusters' - **E.I. Leitinger,** H. Baumgardt, I. Cabrera-Ziri, M. Hilker, E. Pancino
  MNRAS, Volume 520, Issue 1, March 2023, Pages 1456–1480

- 'The kinematics of 30 Milky Way globular clusters and the multiple stellar populations within' - **E.I. Leitinger,** H. Baumgardt, I. Cabrera-Ziri, M. Hilker, J. Carbajo-Hijarrubia, M. Gieles, T.O. Husser, S. Kamman
  Astronomy & Astrophysics, Volume 694, A184, 2025

- 'Not all nitrogen-rich field stars originate from globular clusters' - **E.I. Leitinger,** A. Miglio, J. Montalbán, D. Massari, A. Bragaglia, W.E. van Rossem, K. Brogaard, A. Mazzi, J.S. Thomsen, E.Willett
  Astronomy & Astrophysics, Volume 708, A346, 2026

- 'Not all nitrogen-rich field stars originate from globular clusters II: combining *Kepler* and K2' - **E.I. Leitinger,** E.Pancino, A. Miglio, K. Brogaard, V. Grisoni, D. Alvarez Garay, A. Avdeeva, A. Bragaglia, M. Echeveste, D. Massari, J. Montalbán, S. Nedhath, S. Rani, E. Reggiani, N. Sanna, S. Saracino, L. Steinbauer, A. Turchi
  Astronomy & Astrophysics, submitted

Press releases and research highlights

March 2026	**'The StarXiv' Podcast hosted by Michelle Collins, Payel Das and Nicole Buckley** Episode 32 – Ancient star clusters and growing black holes
May 2026	***Nature Astronomy* research highlight** 'Nitrogen-rich field stars tricking the clock' by Bokyoung Kim DOI: 10.1038/s41550-026-02878-7
June 2026	**SpaCe – Aarhus Space Centre** Press release 1st June 2026: 'Nitrogen-rich field stars look younger than expected' by Ole J. Knudsen
July 2026	**UniBo Magazine** Science article 27th July 2026: 'They appear to be very ancient stars, but their mass makes them appear much younger: what is their true age?' (English translation)
November 2026	**The National Institute for Astrophysics** Press release 11th November 2026: Plato, four programs speak Italian - 'The Milky Way's halo, dated star by star' by Maura Sandri (English translation)

Talks and posters

2026	**Invited talk** PLATO Week	Bologna, Italy
2026	**Contributed talk** MODEST 26 (MOdeling DEnse STellar systems)	Quy Nhon, Vietnam
2026	**Invited talk** APOGEE's Galactic Odyssey	Teruel, Spain
2026	**Contributed talk** TASC10/KASC17: A nod to the past with a view to the future	Aarhus, Denmark
2025	**Poster** Hunting for binaries and exotic stellar populations in star clusters workshop	Florence, Italy
2025	**Contributed talk** Focusing strong lenses: star clusters & globular cluster progenitors at high-z	Sexten, Italy
2025	**Contributed talk & poster** European Astronomical Society (EAS)	Cork, Ireland
2025	**Contributed talk** Stars II - current challenges, upcoming solutions	Bologna, Italy
2024	**Poster** Resolved stellar populations from photographic plates to large surveys	Florence, Italy
2024	**Contributed talk** A lens on globular cluster nurseries: how to compare models with observations	Sexten, Italy
2023	**Invited talk** ESO wine and cheese seminar	ESO, Germany
2023	**Poster** Two in a Million - The interplay between binaries and star clusters	ESO, Germany
2023	**Contributed talk** Star @ Lyon	Lyon, France
2023	**Contributed talk** A multi-wavelength view on GCs near and far: from JWST to the ELT	Sexten, Italy
2022	**Contributed talk** Star-forming clumps and clustered starbursts across cosmic time	MIAPbP, Germany
2022	**Poster** ASA Annual Scientific Meeting	ASA, Australia
2021	**Contributed talk and poster** Mt Stromlo student seminar	Mt Stromlo, Australia
2021	**Poster** UQ SMP poster day	UQ, Australia
2021	**Contributed talk** European Astronomical Society (EAS) Annual Meeting	Virtual
2020	**Contributed talk** 1st Australian BlueMUSE Workshop	Virtual

Organising committees and scientific outreach

2025	**LOC** 'Hunting for binaries and exotic stellar populations in star clusters' workshop	Florence, Italy
2025	**LOC** Haydn mission 1st scientific meeting	Bologna, Italy
2025	**LOC** 'Stars II - current challenges, upcoming solutions' conference	Bologna, Italy
2023	**LOC** 'Two in a Million - The interplay between binaries and star clusters' conference	ESO, Germany
2023	**LOC** 'Coordinated Surveys of the Southern Sky (CSSS)' conference	ESO, Germany
2023	**Organiser** Weekly 'Resolved stellar populations' meeting	ESO, Germany
2023	**Organiser** Weekly 'Galaxy evolution coffee' meeting	ESO, Germany
2023	**Organiser** Weekly 'Informal discussion' meeting	ESO, Germany
2023	**Organiser** Weekly science department social gathering	ESO, Germany
2022	**Organiser/Supervisor** UQ Junior Physics Odyssey (JPhO)	UQ, Australia
2020	**Organiser** Weekly 'Star clusters and galaxies' Meeting	UQ, Australia
2019	**Communications Officer** UQ Physics Club 'PAIN'	UQ, Australia
2019	**Supervisor** Work experience program	UQ, Australia
2018-2019	**Live Science Show Demonstrator** 'Demo Troupe'	UQ, Australia

Scholarships and awards

2022-2023	European Southern Observatory scholarship	ESO, Germany
2022	Winner - ASA Annual Scientific Meeting poster presentation	ASA, Australia
2021	Winner - School of Mathematics and Physics poster day	UQ, Australia
2019-2024	Research training program scholarship	UQ, Australia
2019	Dean's award for academic excellence	UQ, Australia

Observing proposals

PLATO AO-1 Proposal	(**PI**) 'Mapping the accretion history of the Galactic halo with precision stellar ages' *[Accepted]*	2026
LBT 2025B Proposal	(**PI**) 'Ejected or paired? Unveiling the origin of young Nitrogen-rich stars in the Galactic field' *[Accepted] (Program ID IT-2025B-021)*	2025
PLATO AO-1 Proposal	(**Co-I**) 'From detection to inference: a PLATO testbed for globular cluster seismology' *[Accepted]*	2026
TNG AOT53 Proposal	(**Co-I**) 'Helium enrichment from asteroseismology of binary stars' *[Accepted] (Program ID A53TAC_66)*	2025
CAT 2023B Proposal	(**Co-I**) 'Kinematics of multiple populations in NGC5024' *[Accepted] (Program ID 50-GTC46/23B)*	2023
ESO P109 Proposal	(**Co-I**) 'Reticulum - a crucial system to probe the universality of the IMF' *[Accepted] (Program ID 109.23EQ)*	2022

Participation to large projects

Since 2023	**The Wide-field Spectroscopic Telescope (WST)** Science team member	
Since 2024	**Asterochronometry ERC** Lead by Andrea Miglio, Grant Agreement 772293 Post-doctoral researcher: asteroseismology and galactic archaeology	DIFA, Bologna
Since 2025	**PLAnetary Transits and Oscillations of stars (PLATO)** Responsible of Module E: age-kinematics relations in the MSteSci2 pipeline.	
Since 2025	**StarDance ERC** Lead by Elena Pancino, Grant Agreement 101093572 Post-doctoral researcher: globular clusters and binaries	INAF, Florence
Since 2026	**Legacy Survey of Space and Time (LSST)** Junior Associate in the 'Stars, Milky Way, and Local Volume' working group	

Invited institute and group seminar talks

2023	Max Planck Institute for Astronomy seminar	MPA, Garching
2023	European Southern Observatory seminar	ESO, Garching
2025	University of California San Diego seminar	UCSD, California
2026	Leibniz Institute for Astrophysics Potsdam 'Milky Way and Local Volume' research group	AIP, Potsdam

Teaching

2018-2022	**Numerical Methods in Computational Science (COSC2500/7500)** Numerical analysis, linear algebra, solution of equations, and differential equations using Matlab and C++.	UQ, Australia
2019-2022	**Electromagnetism and Modern Physics (PHYS1002)** Electromagnetism, relativity, quantum physics and laboratory experiments.	UQ, Australia
2019-2022	**Physical Basis of Biological Systems (PHYS1171)** Forces and motion, elasticity and fluids, thermodynamics, electricity and biomedical imaging.	UQ, Australia
2019-2022	**Theory and Practice in Science (SCIE1000)** Mathematical and analytical models, introduction to the scientific method, coding in Python and ethics in science.	UQ, Australia
2018-2021	**Physics Laboratory Assistant** Preparation of all undergraduate physics laboratory apparatus throughout each academic semester.	UQ, Australia